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A reinforcement learning approach for optimized MRI sampling with region-specific fidelity

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ABSTRACT

Accelerating Magnetic Resonance Imaging (MRI) acquisition while preserving diagnostic quality remains a significant challenge in medical imaging. This paper introduces a novel reinforcement learning approach for optimizing k-space sampling, addressing the critical need for maintaining high-fidelity reconstructions, particularly in clinically significant regions. Our method bridges k-space and image domains by employing a multi-layer Fast Fourier Transform (FFT) network architecture coupled with a comprehensive reward function. The reward function integrates global SSIM, region-specific metrics, and k-space MSE, achieving SSIM scores of 0.9764 (global) and PSNR of 43.427 in cardiac imaging, with a Dice score of 0.9606 for critical regions. This function uniquely balances global image fidelity with region-specific accuracy, ensuring optimal sampling in areas of high diagnostic value. We present a reinforcement learning framework that maintains consistency with MRI physics throughout the optimization process by utilizing Proximal Policy Optimization (PPO) for concurrent refinement of policy and value networks, while minimizing unnecessary domain transformations. We validate our approach on two diverse datasets: the ACDC cardiac MRI and the FastMRI knee dataset, demonstrating its effectiveness across different regions of clinical interest and showing consistent improvements across 4x and 6x acceleration factors. The proposed k-space sampling optimization technique not only enhances reconstruction quality but also serves as a versatile front-end tool, adaptable to various MRI reconstruction algorithms. The code is publicly available at https://github.com/Ruru-Xu/KSRO

1. Introduction

Magnetic Resonance Imaging (MRI) stands out for its ability to provide detailed, non-invasive visualizations of internal body structures [1]. However, the inherently slow image acquisition process of MRI presents several challenges, including prolonged scan times, patient discomfort, and limited accessibility [2]. These issues have spurred extensive research into accelerated MRI techniques, with k-space sampling optimization emerging as a promising avenue [3,4]. K-space, the raw data domain in MRI, encodes spatial frequency information of the imaged object [5]. Conventional MRI acceleration methods typically involve undersampling k-space and subsequently reconstructing images using various algorithms [6–8]. However, these approaches often struggle to balance global image fidelity with accurate reconstruction of clinically significant regions [9], a critical consideration in diagnostic imaging.

This paper introduces an innovative reinforcement learning approach that bridges k-space and image domains to optimize k-space sampling for accelerated MRI reconstruction. Our method combines a multi-layer Fast Fourier Transform (FFT) network design with a comprehensive reward function that incorporates both global and region-specific quality metrics, as well as k-space fidelity measures. This approach maintains a close connection to the physics of MRI signal acquisition while leveraging image domain information to guide the optimization process, potentially leading to more realistic and artifact-free reconstructions [10,11]. Unlike methods that primarily function in the image domain and require frequent transformations between k-space and image space [12], our approach minimizes domain transformations, enabling more efficient and physically consistent sampling strategies [13]. By operating primarily in k-space while incorporating image domain quality metrics in the reward function, we achieve a balance that ensures both sampling efficiency and clinically relevant reconstruction quality. We demonstrate the effectiveness and versatility of our approach on two distinct datasets: the Automated Cardiac Diagnosis Challenge (ACDC) MRI dataset [14] and the FastMRI knee dataset [15,16]. These datasets represent different anatomical regions and imaging contexts, showcasing the broad applicability of our method. Our key contributions include:

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- A novel network design for k-space sampling optimization, building upon recent advancements in physics-informed neural networks for MRI [17,18].
- A comprehensive reward function that balances global image quality with region-specific accuracy, incorporating both image-domain and k-space metrics. This approach addresses the critical need for targeted reconstruction quality in diagnostically important areas [19].
- A reinforcement learning framework that efficiently bridges k-space and image domains, minimizing unnecessary transformations while maintaining consistency with MRI physics throughout the optimization process [20,21].
- Extensive experiments demonstrating significant improvements in reconstruction quality, particularly in clinically relevant regions, across two datasets and multiple acceleration factors.

By addressing the critical challenge of optimizing k-space sampling through a novel approach that leverages both k-space and image domain information, our work contributes to the broader goal of making high-quality, fast MRI scans more accessible. This has the potential to improve patient care, enhance diagnostic efficiency, and reduce healthcare costs in clinical settings [22,23].

The remainder of this paper is structured as follows: Section 2 reviews related work, Section 3 details our methodology, Section 4 presents our experimental results, and Section 5 concludes with a discussion of implications and future directions.

2. Related work

The field of accelerated MRI reconstruction has seen significant advances in recent years, with approaches broadly categorized as compressed sensing, parallel imaging, and deep learning-based methods. Compressed Sensing (CS) in MRI, introduced by Lustig et al. [24], leverages image sparsity in transform domains to reconstruct high-quality images from undersampled k-space data. However, CS often requires careful parameter tuning and can be computationally intensive. Parallel imaging techniques, such as Sensitivity Encoding (SENSE) [25] and Generalized Auto-calibrating Partially Parallel Acquisition (GRAPPA) [23], utilize spatial information from multiple receiver coils to accelerate image acquisition. However, they are constrained by coil array configurations. The advent of deep learning has revolutionized MRI reconstruction. Convolutional neural networks, particularly U-Net architectures [26], have demonstrated remarkable results in reconstructing images from undersampled data. Schlemper et al. [7] introduced a cascade of Convolutional Neural Networks (CNNs) for dynamic MRI reconstruction, while Qin et al. [27] proposed a convolutional recurrent neural network for improved temporal coherence. Recent advancements include the use of Generative Adversarial Networks (GANs) to enhance the realism of reconstructed images [9]. Emerging techniques in this domain include implicit neural representations [28] and transformer-based models [29], which offer new perspectives on handling complex MRI data structures. Recognizing the importance of preserving details in clinically significant areas, several studies have focused on region-specific reconstruction. Schlemper et al. [30] introduced an attention mechanism to focus on regions of interest during reconstruction. Segmentation-aware MRI reconstruction [31,32] further explored these regions of interest. These approaches highlight the growing emphasis on targeted reconstruction quality in diagnostically important areas. Adaptive sampling in MRI, where sampling patterns are dynamically adjusted based on subject-specific characteristics, represents a cutting-edge approach to acceleration. Gözcü et al. [33] proposed a learning-based framework for optimizing sampling patterns, while Bahadir et al. [34] introduced a method for adaptive sampling using deep reinforcement learning.

Reinforcement Learning (RL) for k-space sampling optimization is a nascent and promising direction. Recent advances in RL convergence and control strategies, including robust distributed control [35], event-triggered consensus mechanisms [36], and nonlinear robust integral-based actor–critic approaches [37], have provided strong theoretical foundations that inform our sampling optimization framework. Yen et al. [38] recently proposed an RL method for MRI pathology prediction, demonstrating the potential of RL in MRI applications, which serves as a foundation for our work. Our work extends this concept, applying RL not for pathology prediction, but for k-space sampling optimization, offering a more flexible framework for accelerated MRI reconstruction.

Our work synthesizes and extends these multiple strands of recent research in MRI reconstruction, introducing novel elements that address key challenges in the field. Specifically, our approach innovates by developing a reinforcement learning-based method for k-space sampling optimization, introducing a comprehensive reward function that balances global image quality with region-specific accuracy, and implementing a reinforcement learning framework that efficiently bridges k-space and image domains while maintaining consistency with MRI physics throughout the optimization process. By focusing on the critical task of k-space sampling optimization and balancing global and region-specific image quality, we aim to contribute to the ongoing efforts to make high-quality, accelerated MRI more accessible and clinically impactful.

3. Methods

Our method for optimizing MRI k-space sampling focuses on three key components. We introduce a novel network architecture integrating FFT convolution [11], U-Net features [26], and an actor–critic framework [38] for effective k-space data processing. Our approach employs a multi-faceted reward function balancing global image quality with critical region accuracy [31]. Finally, we implement a Proximal Policy Optimization (PPO) based learning framework with advanced stability techniques. The following subsections detail each component, explaining how they work together to adaptively learn optimal k-space sampling strategies and improve MRI reconstruction quality, especially in clinically significant areas.

3.1. Network architecture design of the method

Fig. 1 illustrates our integrated pipeline. This method aims to optimize the MRI k-space sampling strategy through reinforcement learning to improve image reconstruction quality. The core of this method is an Actor–Critic network that learns how to select the optimal k-space sampling lines within a limited sampling budget. As an example of the FastMRI knee task, here's a detailed description of the method design process:

- Input Processing: The initial input is a partially sampled k-space data, represented as a 320×320 complex matrix. Three parallel FFT convolution layers process the input with kernel sizes of 9×9, 5×5, and 1×1. Each layer outputs 1 channel. The outputs of these FFT conv layers are concatenated, resulting in a 3-channel feature map.
- UNet Backbone: The concatenated feature map is fed into a UNet architecture. The UNet processes the input and outputs a 64-channel feature map of size 320×320. This step allows the network to capture both local and global features of the k-space data.
- Hybrid Pooling and Feature Extraction: Two types of pooling are applied to the UNet output: (a) Global pooling: AvgPool(32×32) resulting in a $64 \times 32 \times 32$ tensor. (b) Local pooling: AvgPool(64×64) resulting in a $64 \times 64 \times 64$ tensor. Both pooled features are flattened and concatenated, creating a rich feature representation that captures both global and local information.

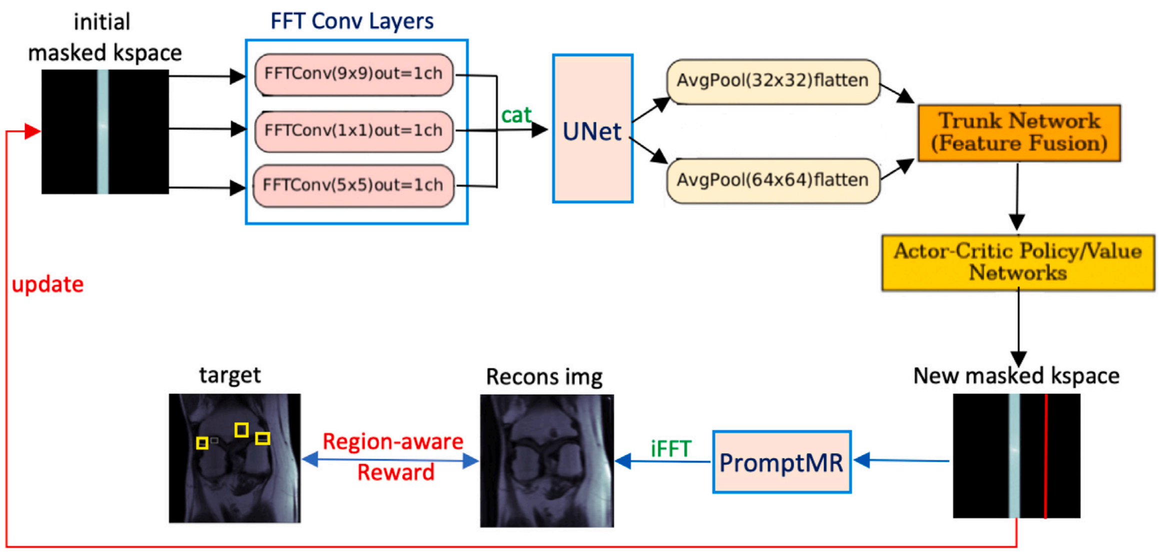

Fig. 1. Reinforcement learning pipeline for optimized k-space sampling in MRI reconstruction. The architecture combines multi-layer FFT convolution, U-Net feature extraction, and an actor–critic network for sampling strategy optimization. It includes a region-aware reward mechanism and utilizes PromptMR for final image refinement, balancing overall image quality and targeted reconstruction of clinically significant regions.

- Trunk Network (Feature Fusion): The concatenated features are passed through a trunk network. This network comprises fully connected layers that further process and fuse the features. The output is a compact feature vector that encapsulates the relevant information from the k-space data.
- Metadata Integration: Additional metadata, representing the number of remaining sampling steps, is one-hot encoded. This encoded metadata is concatenated with the output of the trunk network. This step allows the network to consider the remaining sampling budget in its decision-making process.
- Actor–Critic Policy/Value Networks: The fused features are input to two separate network heads: (a). Policy (Actor) Network: Outputs action logits for 320 possible sampling locations. (b). Value (Critic) Network: Estimates the value of the current state. The policy network uses a masked categorical distribution to ensure only valid sampling locations are considered.
- Action Selection and Environment Interaction: An action (sampling location) is selected based on the policy network's output. The selected action updates the k-space mask, adding a new sampling line.
- Reward Calculation: Then a pre-trained PromptMR [39] model is used to obtain a higher-quality reconstructed image. The reconstructed image is compared to the target image using a region-aware reward mechanism. This reward is based on the improvement in image quality, measured by metrics such as SSIM (Structural Similarity Index).
- Training Loop: The experience (state, action, reward, next state) is collected and used to update the Actor–Critic network. The critic is updated to better estimate state values, while the actor is updated to improve the sampling policy. This process is repeated, gradually refining the sampling strategy.
- Iterative Optimization: The entire process forms a loop, where each iteration provides a new sampling location. As the network learns, it becomes better at selecting sampling locations that maximize image quality improvement.

By considering both the current state of the k-space and the remaining sampling budget, it aims to achieve optimal image reconstruction quality with limited samples. The use of FFT convolutions and the UNet backbone allows the network to effectively process the complex k-space data, while the Actor–Critic architecture enables it to learn a sophisticated sampling policy through trial and error. Our method's ability to dynamically balance global image quality with targeted reconstruction of clinically significant regions ensures overall image enhancement while preserving critical area details.

3.2. Reward functions

Our approach employs a carefully designed reward system that balances overall image quality with precise reconstruction in key diagnostic areas. This method is crucial for maintaining high-quality reconstruction across the entire image while ensuring exceptional detail in critical regions, such as knee lesions or cardiac structures in MRI scans. We have developed three progressively refined reward formulations, each building upon its predecessor to address specific aspects of the reconstruction process.

We start with a basic reward function based on the temporal gradient of the global Structural Similarity Index (SSIM):

$$r_{\text{global}} = \nabla_t \text{SSIM}_{\text{global}} = \text{SSIM}_{\text{global}}(t) - \text{SSIM}_{\text{global}}(t-1) \tag{1}$$

Here, t represents the current time step in the k-space sampling process, and $\text{SSIM}_{\text{global}}(t)$ is the global SSIM value at time step t. This formulation promotes consistent, monotonic improvement in overall image quality throughout the reconstruction process. The sign and magnitude of $r_{\text{global}}(t)$ provide a nuanced signal for the optimization trajectory, guiding the network toward better global reconstruction quality.

Building upon the global measure described above, we introduce a more refined reward function that includes region-specific quality assessment:

$$r = \alpha(t) \cdot r_{\text{global}} + \sum_{i=1}^{N} \beta_i(t) \cdot r_{\text{region}_i} \tag{2}$$

In this equation, $r_{\text{global}}(t)$ is the global SSIM gradient reward, $r_{\text{region}_i}(t)$ is the SSIM gradient reward for the ith region of interest, and N is the total number of regions of interest. The time-dependent weighting coefficients $\alpha(t)$ and $\beta_i(t)$ are dynamically adjusted using an adaptive algorithm that considers the relative SSIM differences between regions. This adaptive scheme enables our model to adjust its focus dynamically during training, optimizing both overall image quality and specific regional details.

The weighting coefficients $\alpha(t)$ and $\beta_i(t)$ are dynamically adjusted using our advanced Algorithm 1:

The algorithm calculates these weights based on several key parameters, where SSIM_i represents the SSIM value for the ith region,

Algorithm 1 Adaptive Multi-Region Weight Calculation

function CALCULATEDYNAMICWEIGHT($\{\text{SSIM}_i\}_{i=0}^{N}$, t, T)
  $\Delta_i \leftarrow |\text{SSIM}_0 - \text{SSIM}_i|$ for $i = 1, \ldots, N$
  $\tau \leftarrow t/T$ // Normalized training progress
  if $\max_i \Delta_i < \epsilon$ **then**
    $\alpha, \{\beta_i\} \leftarrow \frac{1}{N+1}$ // Uniform weights if all SSIMs are similar
  else
    $\alpha \leftarrow \max(0.1, (1-\tau)^2)$ // Quadratic decay for global weight
    $\beta_i \leftarrow \frac{\Delta_i^2}{\sum_{j=1}^{N} \Delta_j^2} \cdot (1-\alpha)$ for $i = 1, \ldots, N$
  end if
  return $\alpha, \{\beta_i\}_{i=1}^{N}$
end function

with SSIM_0 representing the global SSIM. The current time step t and total number of time steps T in the k-space sampling process are used to compute τ, the normalized training progress. A small threshold value ϵ determines SSIM similarity, while Δ_i represents the absolute difference between the global SSIM and the SSIM of the ith region. This adaptive scheme enables our model to adjust its focus dynamically during training, optimizing both overall image quality and specific regional details.

Our final reward formulation combines both spatial and frequency domain data, offering a comprehensive approach to image reconstruction:

$$\begin{aligned} r(t) =& \alpha(t) \cdot r_{\text{global}}(t) + \sum_{i=1}^{N} \beta_i(t) \cdot r_{\text{region}_i}(t) \\ &- \gamma(t) \cdot \sum_{i=1}^{N} \left(\text{MSE}_{\text{k-space},i}(t) - \text{MSE}_{\text{k-space},i}(t-1)\right) \end{aligned} \tag{3}$$

Here, $\text{MSE}_{\text{k-space},i}(t)$ is the Mean Squared Error in k-space for the ith region at time step t, and $\gamma(t)$ is an adaptive weighting factor for k-space fidelity. The function $\gamma(t) = \gamma_0 \cdot \exp(-\lambda t/T)$ balances the importance of k-space fidelity, where γ_0 is the initial weight, λ is a decay factor, and T is the total number of time steps in the k-space sampling process. This exponential decay allows us to gradually shift the emphasis from k-space fidelity to spatial domain quality as the sampling process progresses.

To compute $\text{MSE}_{\text{k-space},i}$, we follow a refined procedure that captures important spectral differences, especially in key diagnostic areas:

$$\text{MSE}_{\text{k-space},i}(t) = \frac{1}{M} \sum_{m=1}^{M} w_m \|\mathcal{F}(x_m(t)) - \mathcal{F}(y_m)\|_2^2 \tag{4}$$

In this equation, M is the number of pixels in the region of interest, w_m is a frequency-dependent weighting factor emphasizing low frequencies, $\mathcal{F}$ denotes the Fourier Transform, $x_m(t)$ is the mth pixel in the reconstructed region of interest at time step t, and y_m is the corresponding pixel in the target region of interest. This approach ensures our model preserves crucial frequency information that might be missed in spatial-only optimization.

Fig. 2 illustrates the comparison between target and reconstructed heart regions, demonstrating how our method assesses MSE in k-space. The top row shows the target heart area in both spatial and k-space representations, while the bottom row displays the reconstructed version. By computing the MSE in this domain, our method effectively captures discrepancies in both coarse and fine image features.

Our reward function evolves from a basic global metric to a detailed spatial-frequency assessment. The whole-image quality gradient provides a foundation for overall improvement, while the region-weighted approach enhances focus on key areas. The integrated reward then adds k-space data to capture fine details critical for diagnosis. This multi-faceted approach allows our model to balance global image quality with precise reconstruction in diagnostically significant regions. The progressive refinement of our reward function offers several advantages. First, it ensures that the overall image quality improves consistently throughout the reconstruction process. Second, it allows for targeted enhancement of critical anatomical structures, which is particularly important in medical imaging. Finally, by incorporating both spatial and frequency domain information, our method can capture subtle details that are crucial for accurate diagnosis but might be overlooked by conventional reconstruction techniques.

3.3. Loss functions

Our optimization strategy leverages the Proximal Policy Optimization (PPO) algorithm, a state-of-the-art reinforcement learning technique. This approach concurrently refines both the policy (actor) and value (critic) networks. The PPO objective function, $\mathcal{L}_{\text{PPO}}$, is a sophisticated composite of several critical components, each addressing specific aspects of the learning process [34]:

$$\mathcal{L}_{\text{PPO}} = \mathbb{E}_t \left[\mathcal{L}_t^{\text{CLIP}}(\theta) - c_1 \mathcal{L}_t^{\text{VF}}(\theta) + c_2 S[\pi_\theta](s_t)\right] \tag{5}$$

where $\mathcal{L}_t^{\text{CLIP}}$ represents the clipped surrogate objective, $\mathcal{L}_t^{\text{VF}}$ is the value function loss, and $S[\pi_\theta](s_t)$ denotes the entropy bonus. The coefficients c_1 and c_2 balance the importance of value function fitting and entropy regularization, respectively.

Policy Gradient Optimization: The core of our policy optimization is the clipped surrogate objective [33]:

$$\mathcal{L}_t^{\text{CLIP}}(\theta) = \hat{\mathbb{E}}_t \left[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t)\right] \tag{6}$$

Here, $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$ is the probability ratio between the new and old policies, and $\hat{A}_t$ is the estimated advantage function. The clipping parameter ϵ acts as a safeguard, constraining policy updates to a trust region and mitigating the risk of overly drastic changes.

Value Function Refinement: For the value function loss, $\mathcal{L}_t^{\text{VF}}$, we employ a clipped objective approach:

$$\begin{aligned} \mathcal{L}_t^{\text{VF}}(\theta) = \frac{1}{2} \max((V_\theta(s_t) - V_t^{\text{target}})^2, \\ (\text{clip}(V_\theta(s_t) - V_{\theta_{\text{old}}}(s_t), -\epsilon, \epsilon) + V_{\theta_{\text{old}}}(s_t) - V_t^{\text{target}})^2) \end{aligned} \tag{7}$$

This formulation incorporates both unclipped and clipped variants, providing a robust mechanism for value estimation while safeguarding against extreme updates. The clipping range is controlled by the same ϵ parameter used in the policy optimization.

Entropy Maximization for Exploration: To promote exploration in the complex k-space sampling domain, we incorporate an entropy maximization term:

$$S[\pi_\theta](s_t) = -\sum_a \pi_\theta(a|s_t) \log \pi_\theta(a|s_t) \tag{8}$$

This term encourages policy diversity, crucial for discovering optimal sampling trajectories in the high-dimensional k-space. The importance of entropy is controlled by the coefficient c_2.

Advanced Optimization Techniques: Our training regimen incorporates several cutting-edge techniques to enhance stability and convergence:

- **Generalized Advantage Estimation (GAE):** We employ GAE to compute advantages, optimizing the bias–variance trade-off in policy gradients:

$$\hat{A}_t^{\text{GAE}(\gamma,\lambda)} = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}^V \tag{9}$$

  where $\delta_t^V = r_t + \gamma V(s_{t+1}) - V(s_t)$ is the temporal difference residual. γ is the discount factor and λ is the GAE parameter.
- **Adaptive KL Divergence Constraint:** We implement an adaptive KL divergence target to maintain policy stability:

$$\text{KL}[\pi_{\theta_{\text{old}}}(\cdot|s_t) \,\|\, \pi_\theta(\cdot|s_t)] \leq \delta_{\text{KL}} \tag{10}$$

  where δ_{KL} is adaptively adjusted based on the observed divergence, which triggers early stopping of policy updates if exceeded.

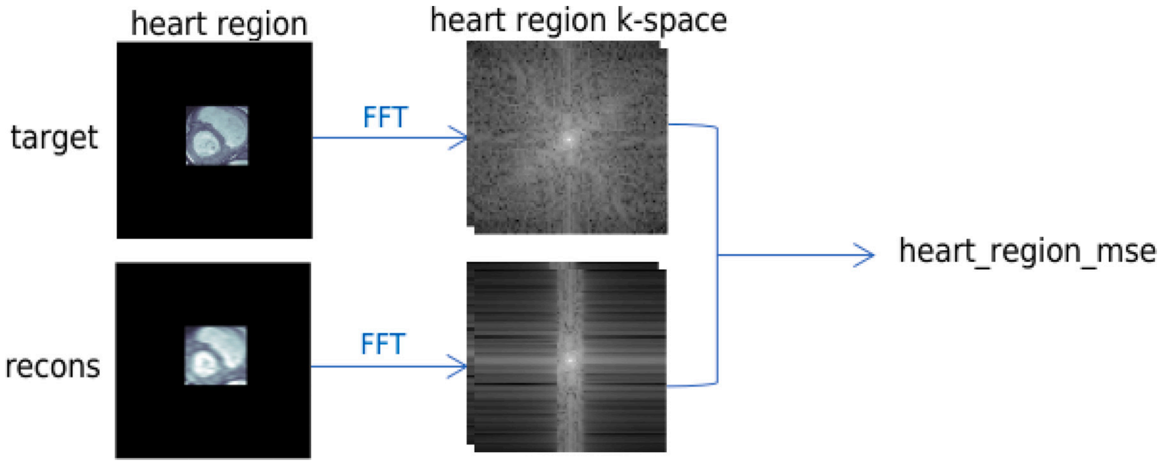

Fig. 2. Target vs. Reconstructed Heart Region Comparison: Top shows the target heart area and its k-space representation. The bottom displays the reconstructed version, highlighting the MSE assessment in k-space.

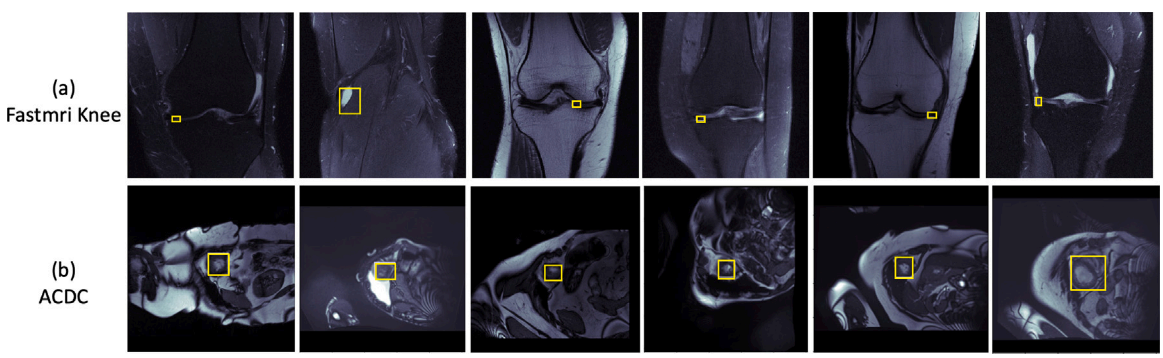

Fig. 3. Sample images from our two study datasets. (a) FastMRI knee scans with yellow boxes highlighting regions of interest, potentially indicating lesions areas. (b) ACDC cardiac MRI slices with yellow boxes emphasizing the heart regions.

By integrating these advanced optimization techniques, our PPO-based framework provides a robust foundation for learning optimal k-space sampling strategies. This approach allows our model to navigate the complex trade-offs between global image quality and region-specific fidelity, adapting dynamically to the unique challenges presented by different MRI reconstruction scenarios.

4. Experimental results

To evaluate the effectiveness of our proposed k-space sampling optimization method, we conducted a comprehensive series of experiments using two different MRI datasets at various acceleration factors. We begin by detailing our experimental setup, including the datasets used and the implementation details of our method. Then, we present a thorough quantitative analysis of our results. This is followed by a qualitative evaluation, where we visually examine the reconstructed images to assess the preservation of clinically relevant features. Finally, we provide an in-depth discussion of our findings, exploring the implications of our results.

4.1. Data preparation

Our study utilizes two distinct MRI datasets to evaluate the effectiveness and versatility of our proposed method: the ACDC dataset [14] for cardiac imaging and the single-coil fastMRI [15,16] knee dataset. These datasets were carefully selected to assess our method's performance across different anatomical regions and clinical applications, providing a comprehensive evaluation of its capabilities in diverse MRI scenarios.

Fig. 3 illustrates representative images from both datasets, highlighting the regions of interest (ROIs) that our method aims to enhance during reconstruction. For the FastMRI knee dataset, these ROIs often correspond to potential lesion areas, while in the ACDC cardiac dataset, they encompass the heart regions.

The ACDC dataset comprises 150 patients, with 1677 slices allocated for training, 367 for validation, and 798 for testing. In contrast, the FastMRI knee dataset is more extensive, encompassing 1172 subjects with 6486 slices for training, 1367 for validation, and 1245 for testing. This substantial dataset size ensures robust training and evaluation of our method.

We implemented a consistent preprocessing pipeline for both datasets, with specific adjustments to account for their unique characteristics. The ACDC images were center-cropped to 256 × 256 pixels, aligning with common deep learning model input sizes, while the FastMRI knee images were cropped to a larger 320 × 320 pixel size to preserve more anatomical detail. Both datasets underwent slice-wise intensity normalization, scaling values to the range [0,1], followed by a Fourier transform to obtain k-space data.

For the ACDC dataset, we augmented the preprocessing by merging non-zero segmentation labels into a single foreground region and computing bounding box coordinates for each slice's foreground. This step facilitates our method's focus on critical cardiac structures. In the case of the FastMRI knee dataset, we extracted single-coil k-space data from each volume and retained pathology labels, enabling our method to potentially correlate sampling strategies with specific knee pathologies.

To ensure data consistency and potentially improve model performance, we computed the global mean and covariance of the k-space data for the respective training sets. We then applied the inverse square root of the covariance matrix to normalize the real and imaginary channels, effectively standardizing the data distribution across both datasets.

All processed data and associated metadata were systematically stored in individual *.h5 files and consolidated into CSV files. This organization ensures efficient data access during both training and subsequent evaluation phases of our method.

Through this comprehensive data preparation process, we have established a robust foundation for training and evaluating our k-space sampling optimization method across diverse MRI applications, setting the stage for a thorough assessment of its effectiveness and generalizability.

4.2. Implementation details

Our experiments were conducted using PyTorch on two high-performance GPUs: an NVIDIA Quadro GV100 with 32 GB memory and an NVIDIA RTX 3090 with 24 GB memory. We implemented our method on both the ACDC and fastMRI knee datasets. For the ACDC dataset, we used 256×256 image and k-space dimensions, with a training batch size of 50 and validation/testing batch sizes of 80. The sampling process involved 32 steps. Our PPO implementation used a batch size of 600, with 6 epochs and 60 mini-batches per update. The fastMRI knee dataset utilized larger 320×320 dimensions, with a sampling range from 0 to 319 and validation/testing batch sizes of 60.

For optimization, we employ the AdamW optimizer with a learning rate of 0.0004 and weight decay of 0.0001, applying gradient clipping with a maximum norm of 0.5. Our PPO hyperparameters included a discount factor of 0.99, GAE lambda of 0.9, clipping coefficient of 0.1, value function coefficient of 0.3, and entropy coefficient of 0.05. We implemented a step-based learning rate scheduler, decreasing the learning rate by a factor of 0.97 every 3000 iterations to improve convergence and performance for training.

4.3. Results analysis

In this study, we evaluated our proposed method on two distinct datasets: the ACDC cardiac MRI dataset and the FastMRI knee dataset, using 4x and 6x acceleration factors. Our experimental design included a range of methods to demonstrate the impact of different reward strategies and network architectures on reconstruction quality.

We begin with a baseline random sampling method to establish a fundamental comparison point. Building upon this, we introduced our first reward-based approach, R1, which utilizes only global SSIM as its reward function. This method serves as an initial benchmark for our reward-based strategies. To evaluate the effectiveness of our network architecture, we included a single-layer FFT approach (1 FFT) as an ablation study, allowing us to illustrate the benefits of our proposed multi-layer design.

Further refining our approach, we developed R2, which incorporates both global and lesion-specific SSIM in its reward function. This method aims to balance overall image quality with targeted improvement in clinically significant regions. Finally, our proposed method, R3, represents the most sophisticated approach, combining global SSIM, lesion-specific SSIM, and k-space MSE in its reward function.

4.3.1. Quantitative evaluation

ACDC Dataset Results:

Table 1 presents the quantitative results for the ACDC dataset. Our key findings include:

- Network Architecture Comparison: Our multi-layer FFT network design (R1) outperformed the single-layer approach (1 FFT) in an ablation study. At 4x acceleration, global SSIM improved from 0.8677 to 0.8771, and lesion SSIM from 0.8012 to 0.8094. This demonstrates the benefit of increased network depth in reconstruction quality.

Table 1
Comparison of reconstruction methods for ACDC Dataset under 4x and 6x acceleration. Metrics include SSIM (Global and Lesion), PSNR (Global and Lesion), and Dice Score. R3 consistently outperforms other methods across all metrics, demonstrating the effectiveness of incorporating lesion-specific information in the reconstruction process.

Method	SSIM		PSNR		Dice score
	Global	Lesion	Global	Lesion	
4x acceleration					
Random	0.7843	0.7355	26.580	24.915	0.7933
R1	0.8771	0.8094	29.932	28.157	0.8904
1 FFT	0.8677	0.8012	29.570	27.790	0.8867
R2	0.8820	0.8164	30.012	28.325	0.8924
R3	**0.8872**	**0.8358**	**30.064**	**29.020**	**0.8982**
6x acceleration					
Random	0.7741	0.7008	26.015	24.236	0.7618
R1	0.8215	0.7386	26.274	25.524	0.8417
1 FFT	0.8085	0.7310	27.190	25.419	0.8175
R2	0.8406	0.7686	28.524	26.727	0.8599
R3	**0.8571**	**0.7921**	**29.254**	**27.476**	**0.8783**

- Reward Strategy Impact: We observed progressive improvements from R1 to R2 and R3. At 4x acceleration, global SSIM increased from 0.8771 (R1) to 0.8820 (R2) and 0.8872 (R3), highlighting the effectiveness of our comprehensive reward function design.
- Resilience Across Acceleration Factors: Our method, particularly R3, demonstrated robustness at higher acceleration factors. At 6x acceleration, R3 maintained a high global SSIM of 0.8571, outperforming other approaches under more challenging conditions.
- Baseline Comparisons: R3 consistently outperformed the random sampling baseline. At 4x acceleration, it improved global SSIM by 13.1% (from 0.7843 to 0.8872) and lesion SSIM by 13.64% (from 0.7355 to 0.8358), underscoring the significant enhancement in reconstruction quality.
- Region-Specific Optimization: The gradual improvement from R1 to R3 emphasizes the importance of incorporating both global and region-specific metrics, as well as frequency domain information. This was particularly evident in lesion-specific metrics and Dice Scores, indicating better preservation of clinically relevant features.

These results demonstrate the effectiveness of our proposed method in enhancing MRI reconstruction quality, particularly in clinically significant regions. The consistent improvement across different metrics and acceleration factors suggests that our approach offers a robust solution for accelerated MRI reconstruction tasks.

FastMRI Knee Dataset Results: Our experimental results on the FastMRI Knee Dataset, as shown in Table 2, demonstrate the effectiveness of our proposed methods. Here are our key findings:

- Progressive Improvement: Across both 4x and 6x acceleration factors, we observed a consistent improvement in reconstruction quality from Random sampling to our most advanced method R3. This trend is evident in both global and lesion-specific metrics.
- Effectiveness of Reward-based Strategies: Our reward-based methods (R1, R2, R3) consistently outperformed random sampling and the single-layer FFT approach. For instance, at 4x acceleration, R3 achieved the highest global SSIM of 0.6805 and lesion SSIM of 0.7381.
- Lesion-specific Enhancement: The improvement in lesion-specific metrics is particularly noteworthy. At 4x acceleration, R3 improved lesion SSIM from 0.6490 (Random) to 0.7381, a significant 13.7% increase. This demonstrates our method's capability to preserve clinically relevant details.
- Robustness to Higher Acceleration: While performance naturally decreased at 6x acceleration, our methods, especially R3, maintained superior performance. R3 achieved a global SSIM of 0.6518

Table 2
Comparison of reconstruction methods for FastMRI Knee Dataset under 4x and 6x acceleration. Metrics include SSIM and PSNR for both global and lesion regions. Methods include Random sampling, Random+ (Random sampling + PromptMR), and three reward-based strategies. R3 consistently achieves the best performance across all metrics, demonstrating the effectiveness of incorporating lesion-specific information in k-space for knee MRI reconstruction.

Method	SSIM		PSNR	
	Global	Lesion	Global	Lesion
4x acceleration				
Random	0.6016	0.6490	23.104	25.314
Random+	0.6268	0.6620	23.566	25.580
R1	0.6738	0.7092	25.987	27.719
1 FFT	0.6619	0.6895	25.627	26.572
R2	0.6777	0.7212	26.070	27.751
R3	**0.6805**	**0.7381**	**26.109**	**28.053**
6x acceleration				
Random	0.5555	0.5992	22.879	24.019
Random+	0.5646	0.6218	23.460	25.669
R1	0.6003	0.6472	24.508	26.461
1 FFT	0.5712	0.6228	23.741	25.924
R2	0.6197	0.6630	24.909	26.758
R3	**0.6518**	**0.6902**	**25.620**	**27.323**

and lesion SSIM of 0.6902 at 6x acceleration, outperforming all other methods.

- Comparison with Post-processing (Random+): Our methods outperformed Random+, which uses random sampling followed by PromptMR post-processing. This indicates that our learning-based sampling strategy is more effective than relying solely on post-processing techniques.
- PSNR Improvements: Consistent with SSIM results, our methods also showed significant improvements in PSNR. At 4x acceleration, R3 achieved the highest global PSNR of 26.109 and lesion PSNR of 28.053.

These results underscore the effectiveness of our proposed method in enhancing MRI reconstruction quality for knee images, particularly in preserving lesion details. The consistent performance across different acceleration factors demonstrates the robustness of our approach in challenging undersampling conditions.

4.3.2. Generalizability and integration with post-processing techniques

For the ACDC dataset, we did not use PromptMR due to the absence of a pre-trained model for cardiac MRI. For the FastMRI knee dataset, we incorporated PromptMR as a post-processing step in all our methods (R1, R2, and R3), similar to the Random+ approach. At 4x acceleration, our R3 method significantly outperformed Random+, achieving a global SSIM of 0.6805 and a lesion SSIM of 0.7381, compared to Random+'s 0.6268 and 0.6620, respectively. This substantial improvement highlights the effectiveness of our intelligent k-space sampling strategy in enhancing the input quality for subsequent reconstruction.

It is important to note that while we used PromptMR in this study, our method is not limited to this specific reconstruction model. Our work can serve as an input enhancement for any reconstruction model, offering a versatile solution for improving MRI reconstruction quality across various applications and anatomical regions. By optimizing the sampling strategy, we provide higher quality inputs for subsequent reconstruction, potentially leading to faster and more accurate diagnostic imaging procedures, regardless of the specific reconstruction model employed.

4.3.3. Qualitative evaluation

ACDC Dataset Visual Analysis: Fig. 4 presents a comprehensive visual comparison of our reconstruction methods applied to the ACDC dataset. The figure includes reconstructed images, global and heart region error maps, and quantitative metrics for each method. Random sampling, our baseline, exhibits the poorest performance, with visible blurring and significant errors across the image, particularly in the heart region. This is reflected in its lowest Dice score of 0.9124. Our proposed methods demonstrate progressive improvements in reconstruction quality:

- R1, incorporating a global SSIM-based reward, shows noticeable enhancement over random sampling, with reduced errors and an increased Dice score of 0.9316.
- The 1 FFT method improves upon random sampling but falls short of our more advanced approaches, supporting our decision to use a multi-layer FFT design.
- R2 and R3, our most sophisticated reward functions, yield the best visual and quantitative results. R3, combining global SSIM, lesion-specific SSIM, and k-space MSE, achieves the highest scores (SSIM_global: 0.9764, PSNR_global: 43.427, Dice score: 0.9606) and produces the cleanest error maps.
- The heart region error maps reveal that R2 and R3 are particularly effective at preserving details in this critical area, reflected in improved SSIM_box and PSNR_box scores.

The visual progression from left to right in the figure demonstrates the incremental improvements achieved by each iteration of our method. This visual evidence, combined with the quantitative metrics, strongly supports the effectiveness of our approach in enhancing MRI reconstruction quality, particularly in clinically significant regions. The consistent improvement in Dice scores further validates the clinical relevance of our method, indicating better alignment with ground truth segmentations.

FastMRI Knee Dataset Visual Analysis: Fig. 5 presents a comparative analysis of our reconstruction methods applied to a knee MRI slice from the FastMRI dataset. This analysis demonstrates the effectiveness of our approach in a different anatomical context from the ACDC dataset. Key observations include:

- Random sampling exhibits significant blurring and loss of detail, particularly in the lesion area, with high-intensity error maps indicating poor reconstruction quality.
- R1, utilizing global SSIM as a reward, shows marked improvement over random sampling, with sharper images and reduced errors in both global and lesion-specific areas.
- The 1 FFT method improves upon random sampling but falls short of our reward-based methods, consistent with our ACDC dataset findings.
- R2 and R3 demonstrate the most significant improvements, with R3 achieving the highest SSIM and PSNR scores (SSIM_global: 0.8449, PSNR_global: 29.765). The error maps for these methods are notably cleaner, especially in the lesion region.
- The lesion region error maps for R2 and R3 show significantly reduced intensity, indicating superior reconstruction quality in this clinically crucial area. This is quantitatively supported by R3's highest SSIM_box (0.8388) and PSNR_box (30.600) scores.
- The visual progression from Random to R3 clearly demonstrates incremental improvements in reconstruction quality, including reduced blurring, enhanced fine details, and improved delineation of anatomical structures, particularly in the lesion area.

The improvements observed in knee MRI reconstruction align with those in cardiac MRI, despite the differences in anatomical structure and imaging characteristics. This consistency underscores the robustness and versatility of our method across different MRI applications. The quantitative metrics align well with the visual improvements, with progressive increases in both global and lesion-specific SSIM and PSNR scores from Random to R3. The substantial improvement in lesion-specific metrics is particularly significant, demonstrating our method's capability to enhance reconstruction quality in diagnostically critical regions.

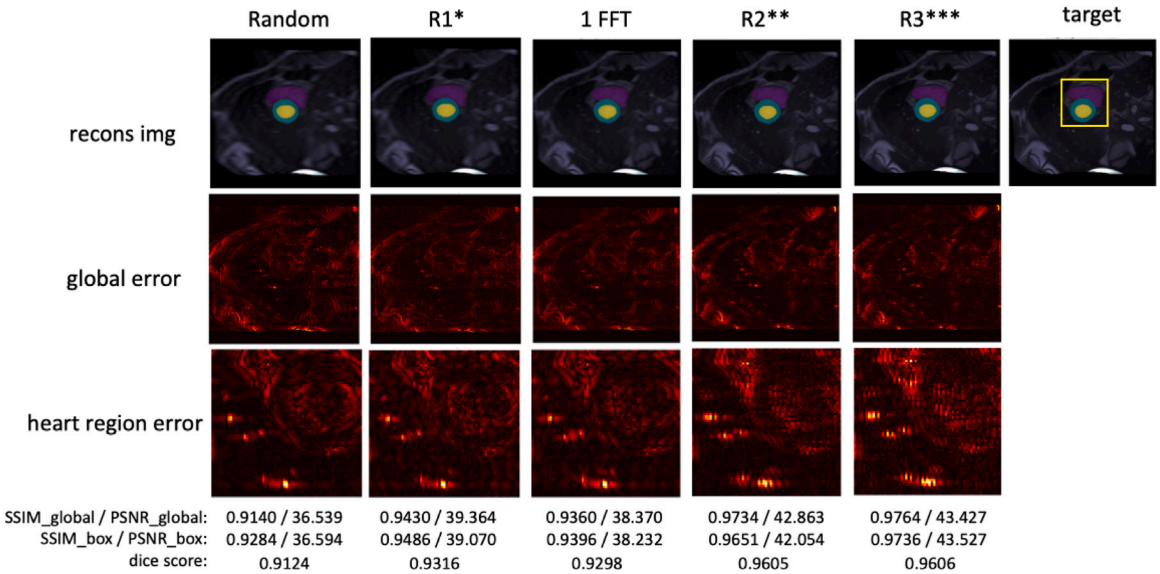

Fig. 4. Comparative analysis of reconstruction methods for the ACDC dataset (4x acc). The figure shows reconstructed images, global error maps, and heart region error maps for each method, along with quantitative SSIM and PSNR scores for both global and heart region (box) areas, and the segmented results.

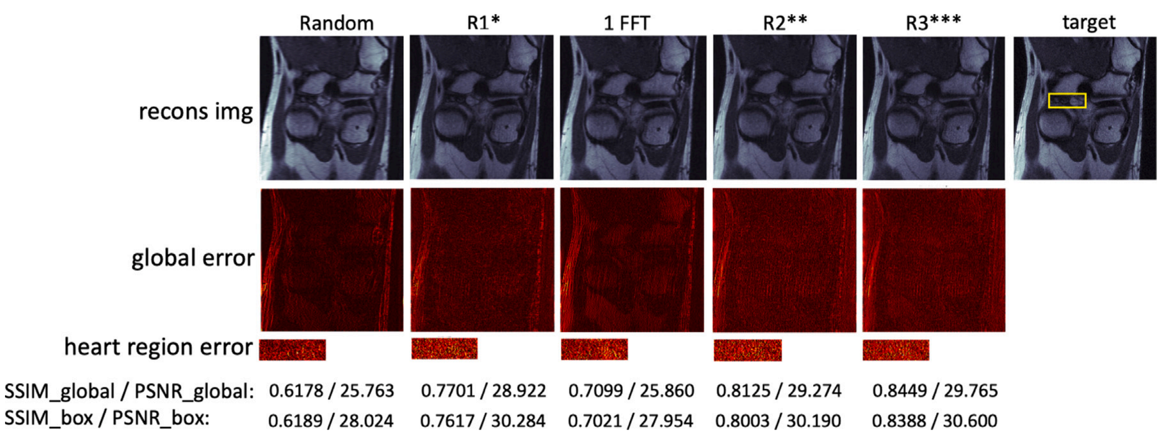

Fig. 5. Comparative analysis of reconstruction methods for the FastMRI Knee dataset (4x acc). The figure displays reconstructed images, global error maps, and lesion region error maps for each method, along with quantitative SSIM and PSNR scores for both global and lesion region (box) areas.

4.3.4. Segmentation performance and clinical relevance

To evaluate the clinical applicability of our reconstruction technique, we employed the nnU-Net [40] pre-trained segmentation model on the ACDC dataset. This approach allows us to assess how reconstruction quality impacts downstream analysis, mimicking a typical clinical workflow of MRI scanning, reconstruction, and automated segmentation.

The segmentation results, quantified by Dice scores, showed consistent improvement across our methods. From the baseline random sampling (Dice score: 0.7933), we observed substantial improvements with R1 (0.8904) and R2 (0.8924) methods. Our most advanced R3 method demonstrated the best performance, achieving a Dice score of 0.8982. This progressive improvement aligns with the increasing sophistication of our reward strategies, providing strong evidence for our approach's effectiveness in preserving clinically relevant anatomical structures. Our R3 method achieved a 13.2% improvement in Dice score compared to random sampling. This significant enhancement, consistent with our observations on SSIM and PSNR metrics, reinforces the robustness of our approach across different evaluation criteria. The consistent outperformance of our R3 method in the nnU-Net segmentation challenge suggests a tangible enhancement in automated analysis accuracy. This indicates that our method preserves detailed information crucial for accelerated MRI protocols, potentially enhancing diagnostic precision in clinical settings.

4.3.5. Comparative analysis of methods

Our experimental results validate the effectiveness of our proposed method across different datasets, acceleration factors, and anatomical regions. The combination of our multi-layer FFT network design and sophisticated reward function (R3) consistently outperforms baseline methods and previous approaches, particularly in clinically significant regions. Key findings from our comparative analysis include:

- Generalizability: Our method demonstrated consistent performance across two distinct datasets – cardiac (ACDC) and knee (FastMRI) – highlighting its robustness and potential for broader application in various MRI contexts.
- Lesion-specific Improvements: Notably, the improvements were particularly pronounced in lesion-specific metrics, which could potentially lead to improved diagnostic accuracy. This is crucial for clinical applications where precise reconstruction of pathological areas is essential.
- Robustness at Higher Acceleration: The widening performance gap between our method and random sampling at higher acceleration factors (6x) demonstrates the robustness of our approach

under more challenging conditions. This is particularly relevant for accelerated MRI protocols in clinical settings.

- Comprehensive Reward Strategy: In both the ACDC and FastMRI knee datasets, our R3 method consistently outperformed other approaches. This comprehensive reward strategy proves effective in guiding the reinforcement learning process to optimize both global image quality and region-specific accuracy.

The visual and quantitative results from both datasets provide strong evidence for the effectiveness of our proposed method in enhancing MRI reconstruction quality. By demonstrating consistent performance across different anatomical regions (cardiac and knee) and imaging characteristics, our method proves its versatility and robustness.

5. Conclusion

In this paper, we have presented an innovative approach to accelerated MRI reconstruction using reinforcement learning for k-space sampling optimization. Our method addresses the challenging balance between overall image quality and precise reconstruction of diagnostically critical regions, a persistent challenge in accelerated MRI techniques.

Our key contributions include the development of a novel multi-layer FFT network operating primarily on k-space data. We developed a sophisticated reward system that uniquely integrates both image domain and k-space domain information. This comprehensive reward function combines global image quality, region-specific accuracy, and k-space mean squared error. This integration ensures a holistic optimization approach that considers both spatial and frequency domain fidelity. Our learning framework efficiently bridges k-space and image domains, minimizing unnecessary transformations while leveraging the strengths of both domains to guide the optimization process. The key strength of our method lies in its ability to optimize k-space sampling patterns, while considering both spatial and frequency domain quality metrics. This approach allows our technique to serve as a front-end for virtually any MRI reconstruction method.

Extensive experimental validation on two distinct datasets (ACDC cardiac MRI and FastMRI knee) demonstrated significant improvements in reconstruction quality across different anatomical regions and imaging contexts. Our results show substantial enhancements in both quantitative metrics (SSIM and PSNR) and visual quality, particularly in regions of high diagnostic value. In conclusion, our reinforcement learning-based approach for k-space sampling optimization, which effectively integrates k-space and image domain information, represents a significant step forward in accelerated MRI reconstruction.

Future research directions include adapting our method for multi-coil MRI systems to enhance performance and broaden applicability, and exploring real-time adaptive sampling during scan acquisition for more efficient and personalized imaging protocols. Additionally, we plan to validate our approach on diverse anatomical regions, including brain and abdominal imaging, across multiple clinical centers to ensure robust generalizability of our method.

CRediT authorship contribution statement

Ruru Xu: Writing – review & editing, Writing – original draft, Methodology. **Ilkay Oksuz:** Writing – review & editing, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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